AI ASSISTENCE CODING- 6.5

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Experiment 6: AI-Based Code Completion: Working with suggestions for classes, loops, conditionals Week3 - Friday

LO1. Use AI-based code completion tools to generate Python code involving classes, loops, and conditionals.

LO2. Interpret and explain AI-generated code line-by-line.

LO3. Identify errors, inefficiencies, or logical flaws in AI-suggested implementations.

LO4. Optimize AI-generated code for better readability and performance.

LO5. Demonstrate ethical and responsible use of AI tools in coding tasks.

Task Description #1

(AI-Based Code Completion for Conditional Eligibility Check)

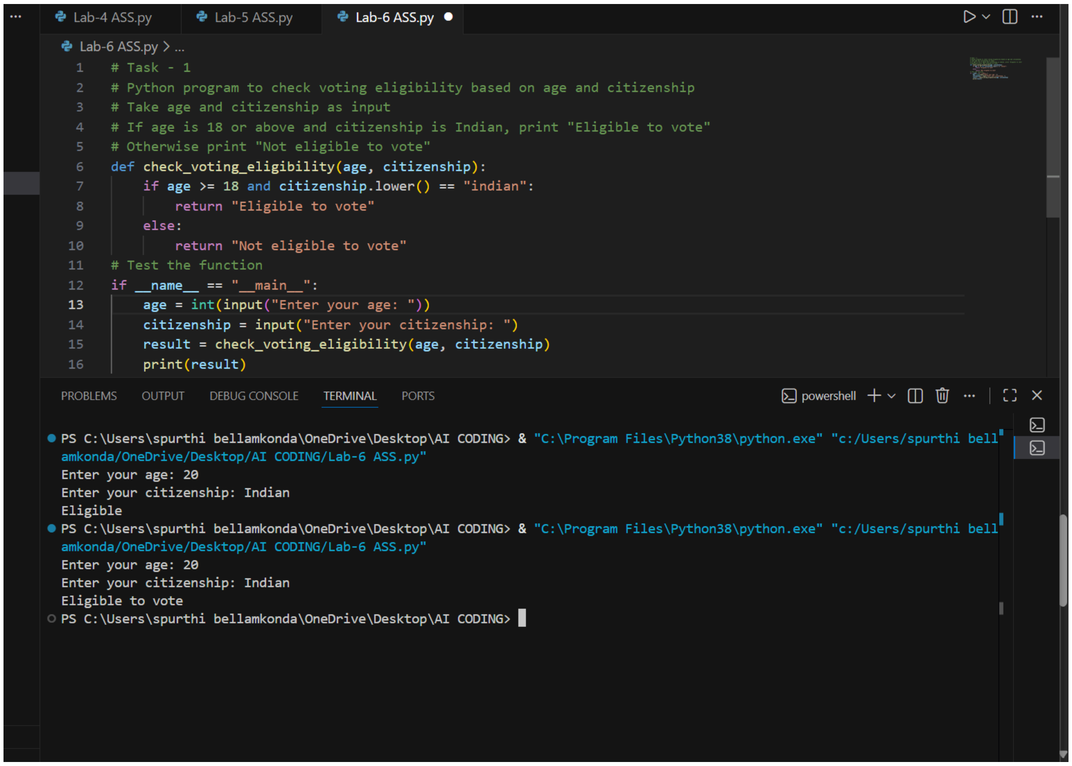
Task: Use an AI tool to generate eligibility logic.

Prompt:

"Generate Python code to check voting eligibility based on age and citizenship."

Expected Output:

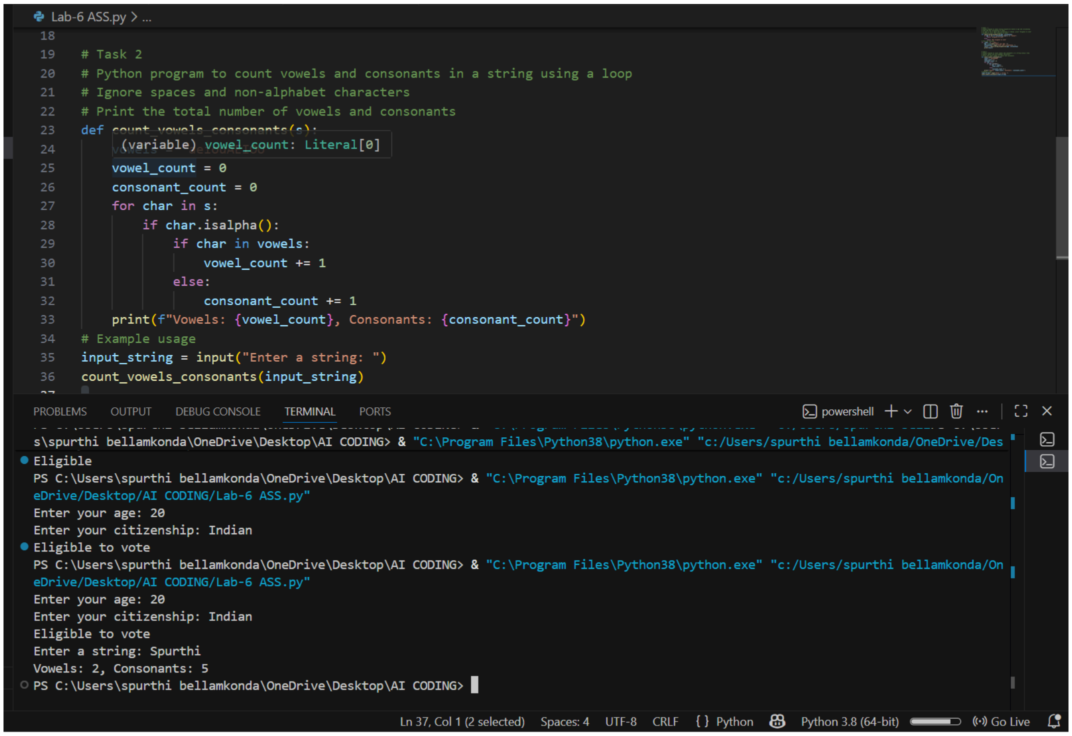
- AI-generated conditional logic.
- Correct eligibility decisions.
- Explanation of conditions.



Task Description #2

(AI-Based Code Completion for Loop-Based
String Processing)
Task: Use an AI tool to process strings using loops.
Prompt:
“Generate Python code to count vowels and consonants in a string
using a loop.”
Expected Output:

• AI-generated string processing logic.
• Correct counts.
• Output verification



Task Description #3

(AI-Assisted Code Completion Reflection
Task)

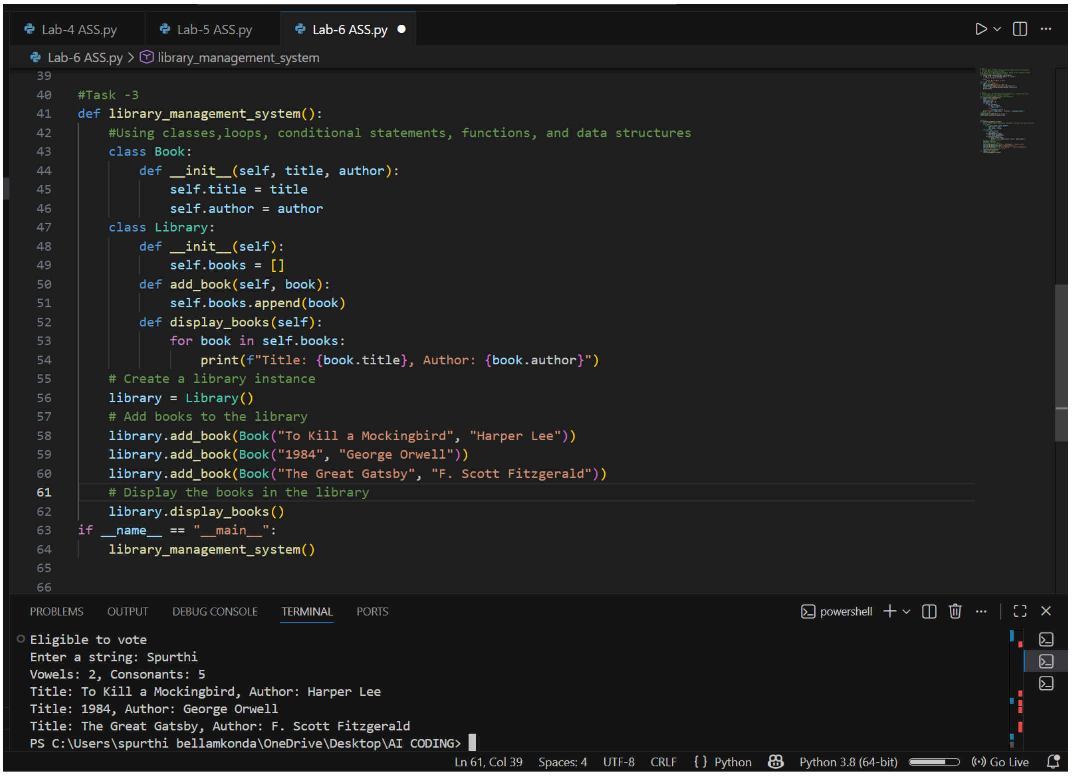
Task: Use an AI tool to generate a complete program using classes, loops, and conditionals.

Prompt:

"Generate a Python program for a library management system using classes, loops, and conditional statements."

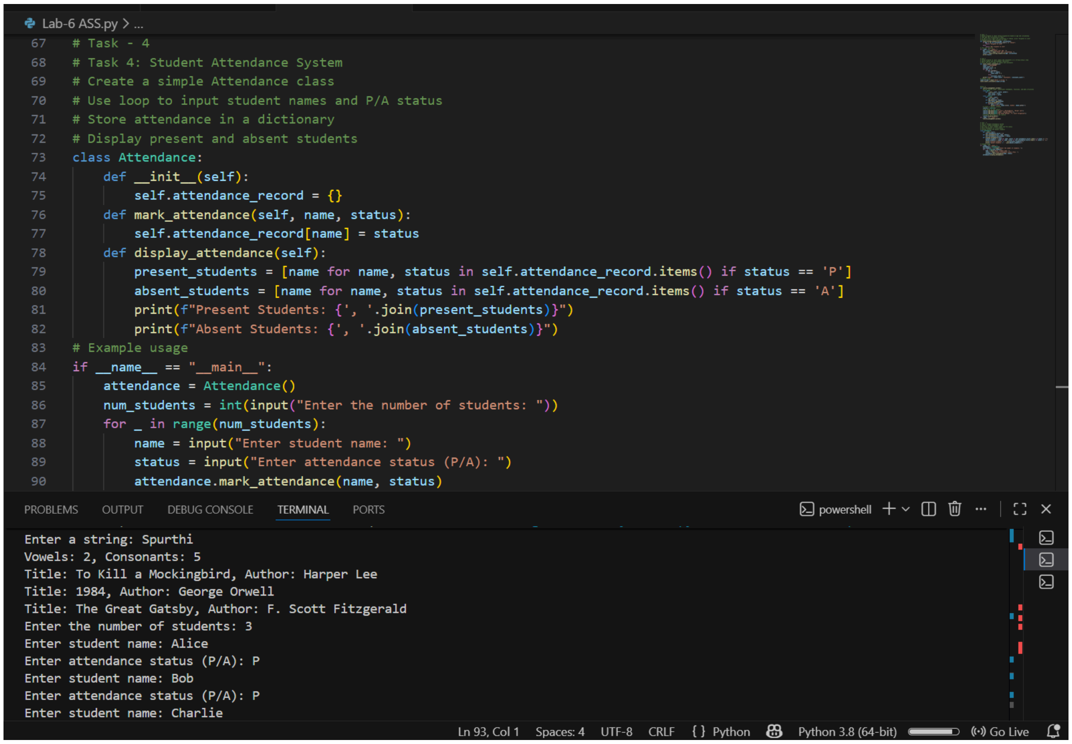
Expected Output:

- Complete AI-generated program.
- Review of AI suggestions quality.
- Short reflection on AI-assisted coding experience.



Task Description #4

(AI-Assisted Code Completion for Class-
Based Attendance System)
Task: Use an AI tool to generate an attendance management class.
Prompt: “Generate a Python class to mark and display student
attendance using loops.”
Expected Output:
• AI-generated attendance logic.
• Correct display of attendance.
• Test cases.



Task Description #5

(AI-Based Code Completion for Conditional
Menu Navigation)
Task: Use an AI tool to complete a navigation menu.
Prompt: “Generate a Python program using loops and conditionals to simulate an ATM menu.”
Expected Output:

- AI-generated menu logic.
- Correct option handling.
- Output verification.

Lab-4 ASS.py | Lab-5 ASS.py | Lab-6 ASS.py

Lab-6 ASS.py > ...

```python
#Task 5
# ATM Menu Simulation
# Initialize balance
# Use while loop to display menu
# Options: check balance, deposit, withdraw, exit
# Use conditionals to handle each option
balance = 1000  # Initial balance
while True:
    print("\nATM Menu:")
    print("1. Check Balance")
    print("2. Deposit Money")
    print("3. Withdraw Money")
    print("4. Exit")
    choice = input("Enter your choice (1-4): ")
    if choice == "1":
        print(f"Your current balance is: ${balance}")
    elif choice == "2":
        amount = float(input("Enter amount to deposit: "))
        balance += amount
        print(f"Deposited ${amount}. New balance is: ${balance}")
    elif choice == "3":
        amount = float(input("Enter amount to withdraw: "))
        if amount <= balance:
            balance -= amount
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell

Lab-4 ASS.py | Lab-5 ASS.py | Lab-6 ASS.py

Lab-6 ASS.py > ...

```python
        amount = float(input("Enter amount to deposit: "))
        balance += amount
        print(f"Deposited ${amount}. New balance is: ${balance}")
    elif choice == "3":
        amount = float(input("Enter amount to withdraw: "))
        if amount <= balance:
            balance -= amount
            print(f"Withdrew ${amount}. New balance is: ${balance}")
        else:
            print("Insufficient funds.")
    elif choice == "4":
        print("Thank you for using the ATM.")
        break
    else:
        print("Invalid choice. Please try again.")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell

```
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkond
eDrive/Desktop/AI CODING/Lab-6 ASS.py"

ATM Menu:
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice (1-4): 3
Enter amount to withdraw: 300
Withdrew $300.0. New balance is: $1200.0

ATM Menu:
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
```